HOL 3: User Management in Ubuntu 22.04

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Learning Objectives

The objective of this Hands-On Lab (HOL) is to help **absolute beginner-level users** understand:

- What user management is in Linux
- Why user management is important in an operating system
- How Ubuntu manages users and groups
- How to perform basic user and group management tasks safely using simple commands

This lab is written in **very easy language**, with **step-by-step explanation**, assuming the learner has **no prior Linux experience**.

Learning Outcomes

After completing this lab, learners will be able to:

- Explain what user management means in Linux
- List users and groups in Ubuntu
- Check which user they are logged in as
- Create users and set passwords
- Understand and check basic permissions
- Create groups and add users to groups
- Perform basic access management tasks confidently

Step 1: What Is User Management in Linux?

User management means controlling:

- Who can log in to the system
- What a user can access
- What actions a user is allowed to perform

Linux is a **multi-user operating system**, which means multiple users can use the same system at the same time without affecting each other.

Step 2: Why Do We Need User Management?

User management is required to:

- Protect the system from unauthorized access
- Separate work between different users
- Improve security
- Control access to files and directories

Example:

- Admin user → full access
- Normal user → limited access

Step 3: Understanding Users and Groups (Very Important)

In Ubuntu:

- Every user belongs to at least one group
- Permissions are decided based on **user**, **group**, and **others**

Important system files:

- `/etc/passwd` → stores user information
- `/etc/group` → stores group information

Step 4: Check Current Logged-in User

Command

```
whoami
```

Explanation

This command shows the username of the currently logged-in user.

Step 5: List All Users in Ubuntu

Command

```
cat /etc/passwd
```

Explanation

This file contains all system and normal users.

List Only Normal Users (Beginner Friendly)

```
awk -F: '$3 >= 1000 {print $1}' /etc/passwd
```

Step 6: List All Groups in Ubuntu

Command

```
cat /etc/group
```

Explanation

This file shows all groups available on the system.

Step 7: Check User and Group Details

Command

```
id username
```

Example:

```
id ubuntuuser
```

Explanation

This command shows:

- User ID (UID)
- Group ID (GID)
- Groups the user belongs to

Step 8: Creating a New User in Ubuntu

Ubuntu provides a **beginner-friendly command** called `adduser`.

Command

```
sudo adduser testuser
```

Explanation

- Creates the user
- Creates home directory
- Asks for password
- Sets default shell

Follow the on-screen instructions.

Step 9: Setting or Changing Password for a User

Command

```
sudo passwd testuser
```

Explanation

This command sets or changes the password for the user.

Step 10: Understanding File Permissions (Basic Level)

Permissions decide who can:

- Read (r)

- Write (w)
- Execute (x)

Check Permissions of a File

```
ls -l file.txt
```

Explanation: Permissions are shown for:

- Owner
- Group
- Others

Step 11: Giving Permissions to a User (Basic Example)

Change file ownership to a user:

```
sudo chown testuser file.txt
```

Give read/write permission:

```
chmod 644 file.txt
```

Step 12: Creating a Group in Ubuntu

Command

```
sudo addgroup devgroup
```

Explanation

This command creates a new group.

Step 13: Adding a User to a Group

Command

```
sudo usermod -aG devgroup testuser
```

Explanation

- `-a` → append (very important)
- `-G` → group name

⚠️Never remove `-a`, otherwise existing groups will be removed.

Step 14: Verify Group Membership

Command

```
id testuser
```

Step 15: Real-Life Beginner Scenario

Scenario: A new employee joins the company.

Steps performed by admin:

1. Create user
2. Set password
3. Add user to team group
4. Give file access

This is exactly what user management is used for in real systems.

End of HOL 3

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